

Simulation study for debiased learning

Simulation setup

Let A be a sample of size $n = 100$ selected from the population U of size $N = 2000$. Let y be a variable of interest and we try to estimate the population total $t_y = \sum_{i \in U} y_i$ and $\mathbf{x}_i$ of size p be the auxiliary variables associated with y where $x_{ij} \stackrel{i.i.d.}{\sim} N(2, 1)$. Suppose the following superpopulation model holds

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2), i = 1, \dots, N$$

where $\sigma^2 = 1$ and $\boldsymbol{\beta} = (1, \dots, 1, 0, \dots, 0)$ with $s = 20$ 1s and $p - s = 20$ 0s. We assume the samples are drawn from the SRS sampling.

We consider the following estimators.

- Difference estimator

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i \right)^T \boldsymbol{\beta} + \sum_{i \in A} d_i (y_i - \mathbf{x}_i^T \boldsymbol{\beta})$$

- GREG estimator

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i \right)^T \hat{\boldsymbol{\beta}} + \sum_{i \in A} d_i (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}})$$

where $\hat{\boldsymbol{\beta}} = (\sum_{i \in A} d_i \mathbf{x}_i \mathbf{x}_i^T)^{-1} (\sum_{i \in A} d_i \mathbf{x}_i y_i)$

- Model-based estimator

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i \right)^T \hat{\boldsymbol{\beta}}$$

- Debiased GREG estimator(Proposed)

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i \right)^T \left(\frac{\sum_{i \in A_2} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_1 + \frac{\sum_{i \in A_1} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_2 \right) + \sum_{i \in A_1} d_i (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_2) + \sum_{i \in A_2} d_i (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1)$$

where A_1 and A_2 are the subsamples of A with $A_1 \cup A_2 = A$, and $\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$ are the regression coefficients built upon the sample A_1 and sample A_2 respectively.

- Design unbiased GREG estimator(Sande, 2021)

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i \right)^T \hat{\boldsymbol{\beta}}_1 + \sum_{i \in A_1} d_i (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1) + \sum_{i \in A_2} \frac{1}{P(i \in A_2 | i \in A)} d_i (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1)$$

```
cbind(BIAS, SE, RMSE)
```

##	BIAS	SE	RMSE
## HT	14.17026931	344.5636	344.8548
## Diff	-0.08421959	118.2858	118.2858
## GREG	218.84918216	198.7317	295.6167

```
## Lasso          161.59757722 181.5888 243.0808
## Debiased       3.79573031 240.6009 240.6309
## Lasso_Debaised 12.38923464 198.2491 198.6358
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.0.5
```

```
ggplot() +
  geom_density(data = data.frame(res), aes(x = Diff, fill = "Diff"), alpha = 0.3) +
  geom_density(data = data.frame(res), aes(x = GREG, fill = "GREG"), alpha = 0.3) +
  geom_density(data = data.frame(res), aes(x = Debiased, fill = "Debiased"), alpha = 0.3) +
  geom_density(data = data.frame(res), aes(x = Lasso, fill = "Lasso"), alpha = 0.3) +
  geom_density(data = data.frame(res), aes(x = Lasso_Debaised, fill = "Lasso_Debaised"), alpha = 0.3) +
  geom_vline(xintercept = t_y, col = "red")
```

